Canonical Projection from the Minimal Carrier

A Proposed Paper AI for the Dream-Paper Suite

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Abstract

Paper AG establishes the minimal carrier $\mathcal{K}^* \hookrightarrow \mathcal{U}$ as the unique minimal sub-object of the universal object through which the canonical transport invariant I_{nl} factors faithfully. Paper AG is explicit that "projection is stronger than persistence": $\mathcal{K}^*$ may contain more observable content than the intended target sector, and the projection residual $\mathrm{Res}(R) = \mathcal{K}^*_R \setminus T_R$ records this gap.

The present paper, Paper AI, proves the Canonical Projection Theorem: within $\mathcal{K}^*$ there is a unique minimal sub-object $T^* \subseteq \mathcal{K}^*$, called the *observable-sufficient target*, through which every stabilized admissible observable of the suite factors faithfully. The canonical projection is the surjective **POT**-morphism $\pi : \mathcal{K}^* \to T^*$ that strips out all content of $\mathcal{K}^*$ that is not observationally accessible.

This result splits Paper E's open burden E7 into two parts: the existence and uniqueness of the canonical projection π (proved here, upgraded to Unc), and the identification of T^* with the specific physical gauge sector of Paper F (which still requires Paper F's realization and structural program, and remains Open).

The key argument is a second-order application of the observable-sufficiency principle: the same philosophy that selected $\mathcal{K}^*$ within $\mathcal{U}$ now selects T^* within $\mathcal{K}^*$. The transport invariant I_{nl} was the tool for the first selection; the full algebra of stabilized admissible observables is the tool for the second.

Contents

1 The projection problem and its split **1**
- 1.1 What Paper AG leaves open . . . 1
- 1.2 The split of E7 . . . 2

2 The observable-sufficiency principle within $\mathcal{K}^*$ **2**

3 The Canonical Projection Theorem **3**

4 Properties of T^* as a POT-object **4**

5 The identification problem: sub-burden E7b **5**

6 Relationship to the NFC Push program's canonicality analysis **5**

7 Status ledger update **6**

8 Conclusion **6**

1 The projection problem and its split

1.1 What Paper AG leaves open

Paper AG proves that there is a unique minimal carrier $\mathcal{K}^* \hookrightarrow \mathcal{U}$ through which the transport invariant I_{nl} factors faithfully. This was the first selection: from all of $\mathcal{U}$, extract the minimal sub-object carrying the invariant.

But I_{nl} measures *transport burden* — observable size, defect count, and collar complexity — not *observable content* directly. $\mathcal{K}^*$ is the minimal sub-object of $\mathcal{U}$ with minimal transport burden. It may still contain:

1. defect-ledger histories that are necessary for I_{nl}'s faithfulness but do not correspond to any specific stabilized admissible observable;
2. collar-type data that bounds the invariant but is not directly observed; and
3. observable equivalence classes that are distinguishable by I_{nl} but correspond to the same underlying admissible test outcome.

The projection problem asks: within $\mathcal{K}^*$, what is the smallest sub-object that carries all *observationally accessible* content — the content that corresponds to stabilized admissible observables rather than merely to transport-invariant values?

1.2 The split of E7

Paper E's open burden E7 is:

> Canonical projection to the target factor.

This splits into two sub-burdens:

(E7a) **Canonical projection exists**: there is a canonical surjective **POT**-morphism $\pi : \mathcal{K}^* \to T^*$ where T^* is the observable-sufficient sub-object of $\mathcal{K}^*$.

(E7b) **Target identification**: T^* coincides with the specific physical gauge sector used as input to Paper F's Yang–Mills bridge program.

Paper AI proves (E7a). Sub-burden (E7b) remains open because it requires Paper F's realization program to identify T^*'s algebraic and geometric character.

2 The observable-sufficiency principle within $\mathcal{K}^*$

Definition 2.1 (Algebra of stabilized admissible observables)**.** The *algebra of stabilized admissible observables* $\mathsf{Obs}^{\mathrm{stab}}$ is the collection of all observational predicates ϕ on $\mathcal{K}^*$ such that:

1. ϕ is defined by an admissible test in the sense of Paper A (it is computed from the behavioral congruence quotient data of $\mathcal{K}^*$);
2. ϕ is stable under the coherent transition maps: $\phi(\tau_\iota(x)) = \phi(x)$ for all admissible enlargements ι and all $x \in \mathcal{K}^*(R)$; and
3. ϕ respects the ORA isomorphisms: $\phi(x) = \phi(\theta(x))$ for all ORA isomorphisms θ from Paper AD.

Remark 2.2 (Relationship to the kernel's observability stance). $\mathsf{Obs}^{\mathrm{stab}}$ is the operationalization, at the level of $\mathcal{K}^* \subseteq \mathcal{U}$, of the flagship kernel's primitive stance: observable structure is defined by admissible tests, and what is real is what survives admissible testing and stabilization. A datum in $\mathcal{K}^*$ that no element of $\mathsf{Obs}^{\mathrm{stab}}$ can detect is, by the suite's own standards, not observable — it is presentation noise that has survived into $\mathcal{K}^*$ via the transport-invariant selection but carries no independent observable meaning.

Definition 2.3 (Observable-sufficient sub-object)**.** A sub-object $T \subseteq \mathcal{K}^*$ is *observable-sufficient* if every $\phi \in \mathsf{Obs}^{\mathrm{stab}}$ factors through T: for every regime R, every $x \in \mathcal{K}^*(R)$, and every $\phi \in \mathsf{Obs}^{\mathrm{stab}}$,

$$\phi(x) \text{ depends only on the image of } x \text{ in } T(R).$$

In other words, T captures all observationally relevant distinctions within $\mathcal{K}^*$.

Definition 2.4 (Observable kernel)**.** The *observable kernel* $\ker_{\mathrm{OS}}$ of the transition $\mathcal{K}^* \to T$ is the collection of equivalence classes of elements of $\mathcal{K}^*$ that are indistinguishable by all of $\mathsf{Obs}^{\mathrm{stab}}$:

$$x \sim_{\mathrm{OS}} y \iff \phi(x) = \phi(y) \text{ for all } \phi \in \mathsf{Obs}^{\mathrm{stab}}.$$

The *observable-sufficient target* is

$$T^* \;=\; \mathcal{K}^*/{\sim_{\mathrm{OS}}}.$$

3 The Canonical Projection Theorem

Lemma 3.1 ($\sim_{\mathrm{OS}}$ is a **POT**-congruence on $\mathcal{K}^*$)**.** *The observable-kernel equivalence relation $\sim_{\mathrm{OS}}$ is compatible with the* **POT***-structure on $\mathcal{K}^*$: for every admissible enlargement ι and* **POT***-transition map τ_ι:*

$$x \sim_{\mathrm{OS}} y \implies \tau_\iota(x) \sim_{\mathrm{OS}} \tau_\iota(y).$$

Proof. Suppose $x \sim_{\mathrm{OS}} y$, meaning $\phi(x) = \phi(y)$ for all $\phi \in \mathsf{Obs}^{\mathrm{stab}}$. By Definition 2.1(2), every $\phi \in \mathsf{Obs}^{\mathrm{stab}}$ is stable under transition maps: $\phi(\tau_\iota(z)) = \phi(z)$ for all z. Therefore

$$\phi(\tau_\iota(x)) = \phi(x) = \phi(y) = \phi(\tau_\iota(y))$$

for all $\phi \in \mathsf{Obs}^{\mathrm{stab}}$, so $\tau_\iota(x) \sim_{\mathrm{OS}} \tau_\iota(y)$. □

Lemma 3.2 ($T^* = \mathcal{K}^*/{\sim_{\mathrm{OS}}}$ is a **POT**-object)**.** *The quotient $T^* = \mathcal{K}^*/{\sim_{\mathrm{OS}}}$ is a well-defined* **POT***-object, and the quotient map $\pi : \mathcal{K}^* \to T^*$ is a surjective* **POT***-morphism.*

Proof. Lemma 3.1 ensures that the transition maps of $\mathcal{K}^*$ descend to well-defined transition maps on the quotient, making T^* a **POT**-object. The boundary-law map descends because $\mathsf{Obs}^{\mathrm{stab}}$ includes collar-outcome observables (any ϕ that detects which collar type is present is in $\mathsf{Obs}^{\mathrm{stab}}$ by Definition 2.1(1)); so the collar-outcome structure on $\mathcal{K}^*$ induces a well-defined collar-outcome structure on T^*.

The quotient map π sends each element to its $\sim_{\mathrm{OS}}$-class. This is a **POT**-morphism by the compatibility of Lemma 3.1, and surjective by definition of quotient. □

Theorem 3.3 (Canonical Projection Theorem)**.** *Let $\mathcal{K}^* \hookrightarrow \mathcal{U}$ be the minimal carrier of Paper AG, and let $T^* = \mathcal{K}^*/{\sim_{\mathrm{OS}}}$ be the observable-sufficient target of Definition 2.4. Then:*

(a) ***Existence****: The canonical surjective* **POT***-morphism $\pi : \mathcal{K}^* \to T^*$ of Lemma 3.2 exists.*

(b) ***Uniqueness of*** T^**: T^* is the unique minimal observable-sufficient sub-object of $\mathcal{K}^*$. Any other observable-sufficient sub-object $T \subseteq \mathcal{K}^*$ satisfies $T^* \hookrightarrow T$.*

(c) ***Route-invariance****: T^* is independent of the enlargement route used to construct $\mathcal{K}^*$: for any two admissible computation paths through* **Reg***, the resulting T^* objects are canonically isomorphic.*

(d) ***Observable completeness****: Every $\phi \in \mathsf{Obs}^{\mathrm{stab}}$ factors through T^*: for every regime R and every $x \in \mathcal{K}^*(R)$, $\phi(x) = \phi'(\pi(x))$ for a unique $\phi' \in \mathsf{Obs}^{\mathrm{stab}}$ defined on $T^*(R)$.*

(e) ***Uniqueness of*** π*: $\pi : \mathcal{K}^* \to T^*$ is the unique surjective* **POT***-morphism satisfying (b)–(d).*

Proof. **Part (a)**: Lemma 3.2.

Part (b) (minimality and uniqueness of T^*): Suppose T is another observable-sufficient sub-object of $\mathcal{K}^*$. Then every $\phi \in \mathsf{Obs}^{\mathrm{stab}}$ factors through T, meaning elements of $\mathcal{K}^*$ that are observationally indistinguishable are identified in T. But $T^* = \mathcal{K}^*/{\sim_{\mathrm{OS}}}$ identifies exactly those elements that are indistinguishable by *all* of $\mathsf{Obs}^{\mathrm{stab}}$, so it makes the coarsest (smallest) quotient among all observable-sufficient ones. Therefore T^* is a quotient of T, equivalently $T^* \hookrightarrow T$ as sub-objects.

Part (c) (route-invariance): $\sim_{\mathrm{OS}}$ is defined intrinsically by $\mathsf{Obs}^{\mathrm{stab}}$, which is defined by the admissible tests of Paper A and their stability conditions. By Definition 2.1(2)–(3), $\mathsf{Obs}^{\mathrm{stab}}$ is invariant under transition maps and ORA isomorphisms. Therefore $T^* = \mathcal{K}^*/{\sim_{\mathrm{OS}}}$ inherits the same route-invariance as the ORA isomorphisms of Paper AD: computing T^* via any two admissible enlargement routes gives canonically isomorphic results.

Part (d) (observable completeness): By definition of $\sim_{\mathrm{OS}}$, elements of $\mathcal{K}^*$ in the same $\sim_{\mathrm{OS}}$-class give the same value for every $\phi \in \mathsf{Obs}^{\mathrm{stab}}$. Therefore each ϕ descends to a well-defined function ϕ' on $T^* = \mathcal{K}^*/{\sim_{\mathrm{OS}}}$, and $\phi(x) = \phi'(\pi(x))$. Uniqueness of ϕ' follows from surjectivity of π.

Part (e) (uniqueness of π): Any other surjective **POT**-morphism $f : \mathcal{K}^* \to T^*$ satisfying (b)–(d) must identify exactly the $\sim_{\mathrm{OS}}$-classes (since (d) requires all observables to descend and (b) requires minimality). So $\ker(f) = \sim_{\mathrm{OS}}$, making $f = \pi$. □

Remark 3.4 (The projection is the second-order observable-sufficiency selection). Theorem 3.3 is a second-order application of the same principle that produced $\mathcal{K}^*$. In Paper AG, the principle was: among all sub-objects of $\mathcal{U}$, take the minimal one through which the invariant I_{nl} factors faithfully. Here the principle is: among all sub-objects of $\mathcal{K}^*$, take the minimal one through which all stabilized admissible observables factor. Both selections are forced by the suite's core operational philosophy: take only what is observationally necessary and nothing more.

4 Properties of T^* as a POT-object

Corollary 4.1 (T^* is a sub-object of $\mathcal{U}$)**.** *$T^* \hookrightarrow \mathcal{K}^* \hookrightarrow \mathcal{U}$ is a chain of* **POT***-sub-objects. T^* is the deepest canonical sub-object of $\mathcal{U}$ that can be specified by the suite's observability doctrine alone.*

Proof. Both inclusions are **POT**-morphisms: the first from Theorem 3.3(a) (viewing π as identifying T^* as a quotient, hence a sub-object of the carrier), the second from Paper AG. The chain is canonical by Theorem 3.3(c). □

Corollary 4.2 (The projection residual is the non-observable content of $\mathcal{K}^*$)**.** *The projection residual $\mathrm{Res}(R) = \mathcal{K}^*_R \setminus T^*_R$ is exactly the set of equivalence classes in $\mathcal{K}^*(R)$ that are:*

1. transport-invariant distinguishable*: distinguishable by I_{nl}, and hence present in $\mathcal{K}^*$ rather than $\mathcal{U} \setminus \mathcal{K}^*$; but*

2. observationally indistinguishable*: not distinguishable by any $\phi \in \mathsf{Obs}^{\mathrm{stab}}$, and hence collapsed in T^*.*

The residual is therefore the collection of defect-ledger and collar-type data that contribute to I_{nl}'s transport accounting but carry no independent observable content.

Proof. By definition: $\mathrm{Res}(R) = \mathcal{K}^*_R \setminus \ker(\pi|_R)$. The kernel of π is $\sim_{\mathrm{OS}}$, which consists of observationally indistinguishable pairs. The elements in Res are those that I_{nl} distinguishes (so they are in $\mathcal{K}^*$, not collapsed to a single class) but $\mathsf{Obs}^{\mathrm{stab}}$ does not (so they are collapsed by π). □

Remark 4.3 (On whether the residual vanishes). Corollary 4.2 characterizes Res precisely but does not determine its size. Whether $\mathrm{Res}(R) = \emptyset$ — whether $T^* = \mathcal{K}^*$ — depends on whether every transport-invariant distinction in $\mathcal{K}^*$ is also an observable distinction. This is a substantive claim about the relationship between I_{nl} and $\mathsf{Obs}^{\mathrm{stab}}$.

In the collar-trivial case, $I_{\mathrm{nl}} = \log|O| + \Lambda$, and the observable content is fully captured by $|O|$ (with Λ providing only transport bookkeeping that is not directly observable). So in general $\mathrm{Res} \neq \emptyset$. The residual is zero precisely when every defect-ledger distinction corresponds to a distinct observable class — a strong condition that may or may not hold for specific regimes.

5 The identification problem: sub-burden E7b

Definition 5.1 (Physical target sector)**.** The *physical target sector* of Paper F is a **POT**-object T_F that:

1. is produced by Paper F's realization program as the persistent canonically projected sector serving as input to Yang–Mills closure; and

2. is characterized algebraically by the commutator/Lie-algebra structure identified in Paper E's sector-extraction program.

Proposition 5.2 (Sub-burden E7b states)**.** *Sub-burden E7b is the claim that $T^* \cong T_F$ as* **POT**-*objects, i.e., the observable-sufficient target selected by Paper AI's canonical projection coincides with the physical target sector of Paper F.*

This is Open*. Establishing it requires:*

1. *showing that T_F is observable-sufficient in the sense of Definition 2.3; and*

2. *showing that T_F is minimal among observable-sufficient sub-objects of $\mathcal{K}^*$ (i.e., $T_F = T^*$ and not merely $T^* \hookrightarrow T_F$).*

Neither of these follows from the present paper. Both depend on Paper F's structural identification of the gauge sector.

Remark 5.3 (What Paper AI achieves compared to E7). Before Paper AI: the projection problem was entirely open (E7 Open).

After Paper AI:

- E7a (canonical projection exists and is unique) is Unc: $\pi : \mathcal{K}^* \to T^*$ is the canonical projection, characterized by observable- sufficiency, route-invariant, and unique.

- E7b (identification $T^* \cong T_F$) remains OPEN: the observable-sufficient target and the physical gauge sector are not yet identified.

The split is the honest outcome. Paper AI proves that the projection exists in the strongest form the suite's observability doctrine supports. Whether the projected object is the physically intended sector requires the downstream realization program.

6 Relationship to the NFC Push program's canonicality analysis

The NFC Push program (Iteration 4) analyzed five candidate canonicality principles and concluded that the strongest, least-smuggled selector is:

> *the unique minimal persistent factor through which all stabilized admissible observables factor faithfully.*

Theorem 3.3 is the formal proof that this principle, applied to $\mathcal{K}^* \subseteq \mathcal{U}$, produces a well-defined, unique, route-invariant target T^*. The five canonicality principles from the Push program map to the theorem's structure:

Push program principle	Theorem correspondence
Route-invariance (Principle R)	Theorem 3.3(c): T^* is route-invariant
Universal-property (Principle U)	Theorem 3.3(b): T^* is the unique minimal observable-sufficient sub-object
Observable-sufficiency (Principle O)	Theorem 3.3(d): all observables factor through T^*
Minimality	Definition 2.4: T^* is the coarsest observable-preserving quotient
Uniqueness	Theorem 3.3(e): π is the unique such morphism

7 Status ledger update

Label	Claim	Status
E7a	Canonical projection $\pi : \mathcal{K}^* \to T^*$ exists and is characterized by universal observable-sufficiency	**Unc** *(proved here)*
AI1	Observable-kernel $\sim_{\mathrm{OS}}$ is a **POT**-congruence	Unc
AI2	$T^* = \mathcal{K}^*/\sim_{\mathrm{OS}}$ is a **POT**-object	Unc
AI3	T^* is the unique minimal observable-sufficient sub-object of $\mathcal{K}^*$	Unc
AI4	π is route-invariant and unique	Unc
AI5	Projection residual = non-observable content of $\mathcal{K}^*$	Unc (characterization)
E7b	Identification $T^* \cong T_F$ with the physical gauge sector	Open
E7	Original combined claim (E7a $\wedge$ E7b)	Partially Unc, partially Open

8 Conclusion

The Canonical Projection Theorem is proved. Within the minimal carrier $\mathcal{K}^* \hookrightarrow \mathcal{U}$, there is a unique minimal sub-object $T^* = \mathcal{K}^*/\sim_{\mathrm{OS}}$, canonically characterized as the quotient of $\mathcal{K}^*$ by observational indistinguishability. The canonical projection $\pi : \mathcal{K}^* \to T^*$ is the unique surjective **POT**-morphism through which all stabilized admissible observables factor. It is route-invariant, minimal, and completely characterized by the suite's own observability doctrine.

The projection residual $\mathrm{Res} = \mathcal{K}^* \setminus T^*$ is the content of $\mathcal{K}^*$ that is transport-invariant distinguishable but observationally indistinguishable: defect-ledger and collar-complexity data that enters the invariant's accounting without corresponding to a direct observable distinction. This residual is now precisely characterized — not hidden, not suppressed.

What remains open is the identification of T^* with the physical gauge sector T_F of Paper F's Yang–Mills bridge (sub-burden E7b). This requires Paper F to show that its extracted finite algebraic sector is precisely the observable-sufficient target of this paper. The connection exists and is natural — Paper F's sector should, by the suite's own consistency, be exactly the minimal observable-sufficient sub-object of the minimal carrier. But that connection is a theorem, not an assertion, and the suite's ledger records it as such.

The suite's canonical layered structure is now:

$$T^* \hookrightarrow \mathcal{K}^* \hookrightarrow \mathcal{U},$$

three canonical sub-objects, each uniquely characterized, with each inclusion defined by a progressively finer observability criterion:

- $\mathcal{K}^* \hookrightarrow \mathcal{U}$: minimal sub-object through which I_{nl} factors faithfully (Paper AG);
- $T^* \hookrightarrow \mathcal{K}^*$: minimal sub-object through which all stabilized admissible observables factor faithfully (Paper AI).

The next paper should close E7b by proving that Paper F's gauge sector is the observable-sufficient target.